

VEERA VENKATA NAGA

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SUMMARY

- Possess over **5+ years** of hands-on experience building and shipping production-grade **full-stack** systems with **React**, **Node.js**, and **Python**, focused on scalable backend and frontend services.
- Adept at designing **RESTful APIs** and **microservices** using **FastAPI**, **Express**, and **Node.js**, delivering reliable, well-tested services that hold up under real production load.
- Experienced in building **AI-powered** applications spanning **RAG** pipelines, **LLM fine-tuning (LoRA)**, **NLP**, and **OCR**, turning unstructured data into usable, source-linked insights.
- Skilled in cloud-native engineering on **AWS** and **Azure**, containerizing services with **Docker** and **Kubernetes** and automating delivery through **CI/CD** pipelines for fast, dependable releases.
- Strong command over **PostgreSQL**, **SQL**, and **ElasticSearch**, with proven **query optimization** across data-heavy systems.

SKILLS

Languages:	Python, JavaScript, TypeScript, SQL, Bash
AI/ML:	LLMs, RAG, Fine-tuning (LoRA), Embeddings, Vector Search, NLP, Human-in-the-Loop, scikit-learn, PyTorch, Hugging Face, pandas, NumPy, OpenCV/OCR
Backend:	Node.js, Express, NestJS, FastAPI, REST APIs, GraphQL, Microservices, WebSockets
Frontend:	React, Redux, Tailwind, HTML5, CSS3, Responsive Design, Jest
Databases:	PostgreSQL, MySQL, MongoDB, Redis, ElasticSearch, Vector DBs
Cloud/DevOps:	AWS, Azure, Docker, Kubernetes, CI/CD, Jenkins, Terraform, MLOps, CloudWatch
CDN/Systems:	CloudFront, Nginx, Caching, Cache Invalidation, Load Balancing, System Design
Practices:	Git, GitHub Actions, Postman, Agile/Scrum, Unit Testing, Code Reviews

EXPERIENCE

Software AI Engineer June 2025 – Present
Hitachi Rail

- Led the design and delivery of an **AI-driven** asset-monitoring service for rail operators, cutting unplanned maintenance downtime by **30%** by turning live sensor data into early failure predictions surfaced through a **React** portal.
- Scaled the **microservices** backend in **Python** and **FastAPI** on **Kubernetes** to ingest telemetry from thousands of train and trackside sensors, sustaining real-time processing as data volume grew **4x**.
- Integrated a **RAG-based** GenAI assistant that let maintenance engineers query manuals and historical work orders in plain language, cutting average troubleshooting lookup time by **40%** across field teams.
- Migrated a legacy monolithic service to cloud-native **microservices** on **AWS**, shrinking release cycles from weeks to days by splitting components into **Docker** containers and automating builds through **CI/CD** pipelines.
- Productionized **machine learning** models through a repeatable **MLOps** workflow, cutting false-positive maintenance alerts by **22%** by automating retraining, monitoring for model drift, and validating each release before rollout.
- Mentored a team of four engineers through code reviews and design sessions, lifting delivery quality and consistency by establishing shared standards for testing, **API** design, and model evaluation across the team.
- Delivered real-time operational dashboards in **React** that gave operators a single view of fleet health, helping crews act on emerging issues up to a day sooner by translating model outputs into prioritized alerts.
- Reduced cloud infrastructure costs by roughly a quarter by profiling workloads, right-sizing **Kubernetes** resources, and caching expensive model inferences, holding latency within the platform's real-time targets.

AI Software Engineer - Graduate Assistant

Jan 2025 – June 2025

University of Illinois Chicago

- Prototyped a **RAG**-based research assistant for a faculty-led study, cutting literature-search time from hours to minutes by grounding an **LLM** in the lab's own corpus through a **Python** backend and lightweight **React** interface.
- Fine-tuned open-source **LLMs** on the lab's domain data using parameter-efficient **LoRA**, lifting answer accuracy over the base model by **18%** through careful prompt design and evaluation against a held-out test set.
- Designed a **human-in-the-loop** review workflow that let researchers correct uncertain model outputs, steadily improving training-data quality by routing low-confidence predictions back to domain experts before each retraining round.
- Benchmarked the **RAG** pipeline against the baseline **LLM** across controlled experiments, sharpening retrieval precision by tuning vector **embeddings**, chunk size, and re-ranking, then documenting results for weekly research reviews.
- Analyzed model behavior and surfaced it through a **React** dashboard, turning raw **LLM** outputs and **eval** metrics into clear, interactive views that shaped the project's direction and fed into a draft research paper.
- Presented findings and live demos to the faculty advisor and a **12-person** research group, securing buy-in for the next phase by translating **RAG** and **fine-tuning** results into plain takeaways the lab could act on.

Software Engineer II

Aug 2022 – Aug 2024

Alstom Transportation

- Architected **iSenS**, an **AI-powered contract management** platform that replaced Alstom's manual review process, cutting average contract review time by **40%** through a **React** workspace backed by **Python** and **FastAPI** services.
- Automated the document-ingestion pipeline that converted scanned, inconsistent contracts into clean structured text using **OCR** with **OpenCV** and **Tesseract** in **Python**, clearing hours of manual data entry across **3,000+** legacy documents.
- Developed the clause-extraction layer with **NLP** that automatically tagged obligations, renewal dates, and parties in each contract, helping reviewers surface risky terms sooner and trimming manual clause review by roughly **30%**.
- Implemented source-linked search over the full contract corpus with **ElasticSearch** and document chunking, cutting the time reviewers spent hunting for the right clause nearly in half by returning ranked, traceable passages.
- Deployed iSenS to production on **AWS** with **FastAPI** services containerized in **Docker**, **S3** storage, and **CloudWatch** logging, keeping the platform stable for daily reviewer use while shipping updates through **CI/CD**.
- Debugged issues spanning the **React** frontend, **REST** API layer, and **PostgreSQL** storage using centralized logs and monitoring, cutting production defects by **25%** and steadily improving reliability release over release.
- Optimized **PostgreSQL** queries and indexing as the contract corpus grew into the tens of thousands, keeping median search response under a second by reworking slow joins and caching the most frequent lookups.
- Collaborated with legal and contract domain experts to shape the platform's **data schemas** and review workflows, lifting clause-extraction accuracy to roughly **90%** by aligning iSenS with how teams reviewed contracts day to day.

Software Engineer

July 2020 – Aug 2022

Ernst & Young

- Drove end-to-end delivery of **full-stack** features across **React** frontends and **Node.js** REST services for client engagements, accelerating feature turnaround by **30%** by streamlining a disciplined biweekly **Agile** release cadence with the team.
- Engineered **RESTful APIs** with **Express** over **SQL** data stores to serve client-facing modules, lifting reliability across **10+** services by standardizing request/response contracts, validation, and consistent error handling across endpoints.
- Overhauled sluggish reporting queries in **SQL** that backed high-traffic client dashboards, cutting page load time by **35%** by rewriting nested joins with **CTEs** and adding targeted indexes on frequently filtered hot columns.

- Consolidated fragmented, duplicated UI into a library of reusable **React** components, speeding feature delivery by **20%** by centralizing state management, prop contracts, and enforcing responsive layout patterns across screens.
- Hardened recurring backend automation written in **Python**, raising job reliability to **92%** by layering in **unit tests**, schema checks, and input validation that caught malformed data before it reached downstream pipeline stages.
- Streamlined releases through **CI/CD** pipelines spanning multiple environments, shrinking deployment cycles by **45%** by automating build, test, and deploy stages on **Git**-based workflows backed by mandatory peer **code reviews**.
- Containerized backend and frontend services with **Docker**, trimming local environment setup time by **50%** by standardizing dependency versions and dev parity across the team's machines and onboarding engineers faster.
- Partnered with product owners and cross-functional stakeholders across **Agile** ceremonies, reducing mid-sprint rework by **25%** by translating ambiguous business requirements into clearly scoped, testable, and documented engineering tickets.

CERTIFICATION

Microsoft Certified: Azure AI Engineer Associate

Feb 2026

Credential ID: 237E973316A0BAC1

- Validated expertise in **generative AI** on **Azure OpenAI** with **prompt engineering** and **RAG** via **Azure AI Search**.
- Proficient in **NLP** and **computer vision** using **Azure AI Language**, **Document Intelligence**, and **OCR** knowledge mining.
- Skilled in securing and monitoring Azure AI workloads via **managed identities**, **Key Vault**, and **Responsible AI** controls.

EDUCATION

University of Illinois Chicago M.S. in Computer Science

Coursework: Distributed Systems, Big Data Visualization, Machine Learning, Data Algorithms, Data Science, AI Engineering

SRM University B.Tech. in Computer Science

Coursework: Software Engineering, Data Structures, Operating Systems, Network Communication, Database Management

PROJECTS

RAG-Powered Research Assistant | *Python, FastAPI, Hugging Face, Vector Search, React*

- Built a **RAG** pipeline grounding an **LLM** in a custom corpus using **embeddings**, chunking, and **vector search**.
- Served it via a **FastAPI** backend and **React** UI, returning source-cited, **evaluation**-validated answers.

Fine-Tuned Transformer for Text Classification | *Python, PyTorch, Hugging Face, scikit-learn, pandas*

- Fine-tuned a transformer with **LoRA** on a labeled dataset, reaching **92%** F1 over **scikit-learn** baselines.
- Tracked experiments and **model evaluation** metrics with **pandas**, comparing accuracy, F1, and latency across sizes.

Distributed Microservices Platform | *Node.js, NestJS, REST, Docker, Kubernetes, Redis*

- Designed fault-tolerant **microservices** with **REST APIs** and **Redis** caching, using retries, timeouts, and health checks.
- Deployed the services on **Kubernetes** with autoscaling and **Docker**, holding low latency under high-concurrency load.

Cloud-Native Data Pipeline | *Python, AWS, Docker, CI/CD, PostgreSQL*

- Built a cloud-native data pipeline transforming **10K+** records through validated, scheduled runs into **PostgreSQL**.
- Containerized each stage with **Docker** and automated build, test, and deployment through **CI/CD** on **AWS**.

Full-Stack Analytics Dashboard | *React, Node.js, Express, PostgreSQL, Docker*

- Built a **full-stack** dashboard with a **React** frontend and **Node.js/Express** REST APIs over **PostgreSQL**.
- Added **JWT** authentication and role-based access, then containerized the app with **Docker**.